

VPC USING BASTION SERVER

1. VPC with Bastion Server

VPC & Subnet Design

VPC CIDR: 10.0.0.0/16

Public Subnet (Bastion Server): 10.0.1.0/24

Private Subnet (Database Server): 10.0.2.0/24

2. Security Group Configuration (VERY IMPORTANT)

Bastion Server Security Group (Public Subnet)

Allows SSH (22) from: My IP (Database Administrator)

Database Server Security Group (Private Subnet)

Allows MySQL / Aurora (3306) from: custom 10.0.1.0/24 (Public subnet / Bastion subnet)

Allows SSH (22) from: custom Private IP of Bastion Server

Does NOT allow public access

3. Security Outcome

Database server is NOT publicly accessible

Only Bastion server can connect to DB server using SSH

External users cannot access DB directly

4. EC2 Instance Creation

Bastion Server

Subnet: Public Subnet (10.0.1.0/24)

Key pair: bastion.pem

Operating System: Ubuntu

Database Server

Subnet: Private Subnet (10.0.2.0/24)

Key pair: dbserver.pem

Operating System: Ubuntu

5. Copy DB Server Key to Bastion Server

Database server is in private subnet

Direct SSH from local system is NOT possible

DB server key must be copied to Bastion server

Command (PowerShell):

```
scp -i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
```

File is copied securely using SCP

dbserver.pem will now be available inside Bastion server

6. Connect to Bastion Server

SSH Command:

```
ssh -i bastion.pem ubuntu@<BASTION_PUBLIC_IP>
```

Verify key files:

```
ls
```

Output:

```
bastion.pem
```

dbserver.pem

7. Connect from Bastion to Database Server

SSH from Bastion:

```
ssh -i dbserver.pem ubuntu@10.0.2.x
```

Connection to DB server is successful

Database administrator can now:

Update database configurations

Install required packages-----problem occurs

Restart MySQL service

VPC USING NAT GATEWAY

8. In the above steps it clear there is no internet access, if we want to install or update anything in the instances present in private subnet.
9. Network Address Translator (NAT Gateway) will solve this problem. Goto NAT Gateway in the left side menu bar of VPC and click on create NAT gateway.
10. Give the Nat gateway name, select your VPC, scroll down and click create NAT gateway.
11. So Now create a route for accessing NATGateway. Goto Route table click on create route table and give the route table name as bastion route , select our customized vpc and click on create route.
12. Select the route, click on actions, goto edit subnet associations and add private subnet to this bastion route.
13. Again goto actions and click on edit routes, attach NAT Gateway with 0.0.0.0/0, select your NAT gateway name in the drop down menu and click on save changes.
14. Now check in the dbserver you will internet in ec2 of private subnet.

SCENARIO BASED QUESTIONS

1. You have multiple private subnets in different AZs. Should you use **one NAT Gateway or multiple NAT Gateways**? Explain your design choice.
2. Your private subnet instances can reach the internet, but the NAT Gateway shows no traffic. What could be the possible reasons?
3. A NAT Gateway was accidentally created in a **private subnet** instead of a public subnet. What problem will occur, and how would you fix it?
4. An application running in a private subnet requires outbound access only to a few trusted external services.
How would you restrict traffic while still using a NAT Gateway?
5. Multiple teams share a single VPC and NAT Gateway. One team's traffic is affecting others. How would you redesign the NAT Gateway setup to isolate traffic?
6. You can successfully SSH into a private EC2 instance using a Bastion server, but yum update / apt update fails inside the private instance. Why?
7. Private EC2 instances can access the internet through a NAT Gateway, but administrators cannot SSH into them using the Bastion server.
8. A Bastion server and NAT Gateway are both created in a **private subnet** by mistake. What impact does this cause?
9. SSH access to the Bastion server works, but SSH from Bastion to the private EC2 fails. Why?

10. A student claims that a Bastion server alone is enough to provide internet access to private instances. Why is Bastion not a replacement for NAT Gateway?